FUN-3.D.1 The chain rule is the basis for implicit differentiation.

3-2 Implicit Differentiation

Generally, we have been working with functions written in **EXPLICIT FORM**, meaning that y is expressed as a function of x. However, some functions are difficult or impossible to write in explicit form. Instead, they are left in **IMPLICIT FORM**, meaning that x and y are written together in an expression without solving for a single variable.

Explicit Form	Implicit Form
$y = 2x^2 - \sin(x) + 4$	$3x^2 - 2y^3 + 4xy = 7$

Key Question: What does it mean to "find a derivative"?

This question is often misunderstood because we have only been working with explicitly written functions so far. If given an equation involving y, asking to find the derivative is equivalent to finding or solving for $\frac{dy}{dx}$.

Reminder: The derivative of y is $\frac{dy}{dx}$. Writing $\frac{dy}{dx}$ indicates that we are taking the derivative of y with respect to x. Taking the derivative of an expression involving y requires the same rules that we used for expressions with x. Derivatives require the **chain rule**!

Example 1: Differentiate the following with respect to x

y is a function of x

a. $\frac{d}{dx}[x^3] = 3x^2$

b. $\frac{d}{dx}[y^3] = 3y^2 \cdot \frac{dy}{dx}$

c. $\frac{d}{dx}[x + 3y] = 1 + 3 \cdot \frac{dy}{dx}$

d. $\frac{d}{dx}[xy^2]$

$= 1 \cdot y^2 + x \cdot 2y \cdot \frac{dy}{dx}$

Guidelines for Implicit Differentiation
Step 1: Differentiate both sides of the equation with respect to x. Each time the derivative of a "y-term" is involved, include a $\frac{dy}{dx}$.
Step 2: Collect all terms involving $\frac{dy}{dx}$ on the left side of the equation and everything else on the right.
Step 3: Factor out $\frac{dy}{dx}$ on the left side, if necessary, to create only one $\frac{dy}{dx}$ term.
Step 4: Solve for $\frac{dy}{dx}$.

Example 2: Find $\frac{dy}{dx}$ given that $y^2 + 2xy - x^3 = 4$

$$2y \cdot \frac{dy}{dx} + 2y + 2x \cdot \frac{dy}{dx} - 3x^2 = 0$$

$$2y\frac{dy}{dx} + 2x\frac{dy}{dx} = -2y + 3x^2$$

$$\frac{dy}{dx}(2y + 2x) = -2y + 3x^2$$

$$\frac{dy}{dx} = \frac{-2y + 3x^2}{2y + 2x}$$

OR $\frac{3x^2 - 2y}{2y + 2x}$ OR $\frac{2y - 3x^2}{-2y - 2x}$

Example 3: Find $\frac{dy}{dx}$ given $x^3 - xy + y^3 = 0$

$$3x^2 - 1 \cdot y - x \cdot \frac{dy}{dx} + 3y^2 \cdot \frac{dy}{dx} = 0$$

$$-x\frac{dy}{dx} + 3y^2\frac{dy}{dx} = -3x^2 + y$$

$$\frac{dy}{dx}\left(-x + 3y^2\right) = -3x^2 + y$$

$$\frac{dy}{dx} = \frac{-3x^2 + y}{-x + 3y^2}$$

Example 4: Write an equation of the line tangent to the graph of $x^2 - y^3 = 0$ through the point $(1, 1)$.

$$2x - 3y^2 \cdot \frac{dy}{dx} = 0$$

Plug in $(1,1)$: $2 - 3\frac{dy}{dx} = 0$

$$-3\frac{dy}{dx} = -2$$

$$\frac{dy}{dx} = \frac{2}{3}$$

$\left.\frac{dy}{dx}\right|_{(1,1)} = ?$ Slope

$\left.\frac{dy}{dx}\right|_{(1,1)} = \frac{2}{3}$

EQ_{TAN}:

$$y - 1 = \frac{2}{3}(x - 1)$$

Example 5: Find y' if $\sin(x+y) = y^2 \cos x$.

$$\cos(x+y)\cdot\left(1+\frac{dy}{dx}\right) = 2y\frac{dy}{dx}\cos x + y^2(-\sin x)$$

$$\cos(x+y) + \cos(x+y)\frac{dy}{dx} = 2y\frac{dy}{dx}\cos x - y^2\sin x$$

$$\cos(x+y)\frac{dy}{dx} - 2y\frac{dy}{dx}\cos x = -\cos(x+y) - y^2\sin x$$

$$\frac{dy}{dx}\left[\cos(x+y) - 2y\cos x\right] = -\cos(x+y) - y^2\sin x$$

$$\frac{dy}{dx} = \frac{-\cos(x+y) - y^2\sin x}{\cos(x+y) - 2y\cos x}$$

Example 6: Consider the curve given by $xy^2 - x^3y = 6$.

a. Show that $\frac{dy}{dx} = \frac{3x^2y - y^2}{2xy - x^3}$

b. Find all points on the curve whose x-coordinate is 1 and write an equation for the tangent line at each of these points.

Horizontal and Vertical Tangent Lines	Horizontal tangent lines exist when the slope $\frac{dy}{dx} =$ Vertical tangent lines exist when the slope $\frac{dy}{dx} =$

Example 7: Given the curve $3x^2 + 2y^2 = 16$

a. Find the coordinates of the points on the curve that have horizontal tangent lines.

b. Find the coordinates of the points on the curve that have vertical tangent lines.